AEROSPACE STANDARD

AS81934™	REV. C
Issued 1998-04 Revised 2019-09 Superseding AS81934B	

(R) Bearings, Sleeve, Plain and Flanged, Self-Lubricating

RATIONALE

This document includes limited scope changes to the following sections in response to comments from the approved Rev. C ballot (April 2019): 3.6.2, 4.3.1, 4.3.3, 4.6.6.2, and Figure 1. Previously approved changes from Rev. B to Rev. C: Updating the high/sub-zero temperature test methods, room temperature break-in cycles, qualification requirements by similarity to AS81820 for CRES substrates, and test pin and housing tolerances. Adding/updating fluid contaminants and specifying type and concentration of anti-icing fluid to be used for testing. Adding retention of qualification requirements for aluminum substrate. Clarifying tables and text to address conflicting requirements for number of qualification test samples required. Clarifying when LN_2 exposure is required.

NOTICE

The initial SAE publication of this document was taken directly from U.S. Military Standard MIL-B-81934B. This SAE Standard may retain the same part numbers established by the original military document.

1. SCOPE

1.1 Scope

This standard covers plain and flanged sleeve bearings which are self-lubricating by incorporating polytetrafluoroethylene (PTFE) in a liner in the bore for use in the temperature range of -65 to +325 °F.

2. REFERENCES

2.1 Applicable Documents

The following publications form a part of this document to the extent specified herein. The latest issue of SAE publications shall apply. The applicable issue of other publications shall be the issue in effect on the date of the purchase order. In the event of conflict between the text of this document and references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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2.1.1 SAE Publications

Available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, Tel: 877-606-7323 (inside USA and Canada) or +1 724-776-4970 (outside USA), www.sae.org.

AMS1424	Fluid, Aircraft Deicing/Anti-Icing, SAE Type 1
AMS2417	Plating, Zinc-Nickel Alloy
AMS2700	Passivation of Corrosion Resistant Steels
AMS-QQ-P-416	Plating, Cadmium (Electrodeposited)
ARP5448/9	Plain Bearing Lined Inside Diameter Plug Gaging
AS1241	Fire Resistant Phosphate Ester Hydraulic Fluid for Aircraft
AS14101	Bearing, Plain, Self-Lubricating, Self-Aligning, Low Speed, Narrow, Grooved Race, -65 to +325 °F
AS81820	Bearings, Plain, Self-Aligning, Self-Lubricating, Low Speed Oscillation
AS81934/1	Bearing, Sleeve, Plain, Self-Lubricating, -65 to +325 °F
AS81934/2	Bearing, Sleeve, Flanged, Self-Lubricating, -65 to +325 °F

2.1.2 ASTM Publications

Available from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959, Tel: 610-832-9585, www.astm.org.

ASTM A967	Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts
ASTM C794	Standard Test Method for Adhesion-in-Peel of Elastomeric Joint Sealants
ASTM D1655	Standard Specification for Aviation Turbine Fuels
ASTM D3951	Standard Practice for Commercial Packaging
ASTM F25	Standard Test Method for Sizing and Counting Airborne Particulate Contamination in Clean Rooms and Other Dust-Controlled Areas
ASTM F50	Standard Practice for Continuous Sizing and Counting of Airborne Particles in Dust-Controlled Areas and Clean Rooms Using Instruments Capable of Detecting Single Sub-Micrometre and Larger Particles

2.1.3 ASME Publications

Available from ASME, P.O. Box 2900, 22 Law Drive, Fairfield, NJ 07007-2900, Tel: 800-843-2763 (U.S./Canada), 001-800-843-2763 (Mexico), 973-882-1170 (outside North America), www.asme.org.

ASME B46.1	Surface Texture (Surface Roughness, Waviness and Lay)
ASME Y14.100	Engineering Drawing Practices

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2.1.4 U.S. Government Publications

Copies of these documents are available online at https://quicksearch.dla.mil.

MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-DTL-197	Packaging of Bearings, Associated Parts and SubAssemblies
MIL-DTL-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-DTL-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-83133	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
MIL-HDBK-1599	Bearings, Control System Components and Associated Hardware Used in the Design and Construction of Aerospace Mechanical Systems and Subsystems.
MIL-PRF-5606	Hydraulic Fluid, Petroleum Base; Aircraft; Missile and Ordnance
MIL-PRF-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-PRF-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
MIL-STD-2073-1	Standard Practice for Military Packaging

2.2 Definitions

2.2.1 PROCESSING PROCEDURES

All bonding, curing, and postcuring procedures.

2.2.2 VOID

A void is an area where the metal substrate is directly visible with no visible adhesive after peeling the liner.

2.2.3 UNBONDED AREA

An unbonded area is where the adhesive remaining on the metal substrate is smooth and shiny indicating a lack of bonding pressure.

3. REQUIREMENTS

3.1 Aerospace Standard (AS) Sheets

The individual item requirements shall be as specified herein and in accordance with the applicable AS sheets. In the event of any conflict between the requirements of this standard and the AS sheet, the latter shall govern.

3.2 Qualification

Products furnished under this standard shall be products which are authorized by the qualifying activity for listing on the applicable Qualified Products List (QPL-AS81934) at the time of award of contract (see 4.3, 6.3, and 6.3.1). See Qualified Products Database (QPD) at https://assist.dla.mil/, https://qpldocs.dla.mil/, or https://quicksearch.dla.mil/.

3.2.1 Product Change

Any change in product design, description, materials, processing procedures, or plant location shall be reported to the qualifying activity and may require requalification of the product to an extent determined by the qualifying activity. Any other specific changes, which must be brought to the qualifying activity's attention, shall be identified in the qualification notification letter.

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3.2.2 Product Manufacture

Except for the bonding operations, the manufacturer is permitted to subcontract manufacturing operations without violating the requirements of 3.2.1. The bonding operations shall be performed in the plant listed on the Qualified Products List. Manufacture of the self-lubricating liner material may also be subcontracted. Any change in (1) the liner manufacturer, (2) the liner manufacturing procedures, or (3) the materials used in manufacture of the liner will require requalification to an extent determined by the qualifying activity. The manufacturer is responsible for meeting all requirements of the standard and for the quality of the end product, whether it is manufactured totally in-house or some of the operations are performed by a subcontractor. Inherent in the responsibility for the end product is the responsibility to verify that the subcontractor's processes meet standard requirements.

3.3 Materials

Material for the sleeve and liner shall conform to the applicable AS sheet. PTFE shall be included in the liner in such a manner that the bearing will conform to all requirements of this standard.

3.3.1 Corrosion Resistant Steel Finishes

Corrosion resistant steel (CRES) sleeve bearings shall be finished by cadmium plating, zinc-nickel plating, or passivating in accordance with 3.3.1.1 or 3.3.1.2, and as defined with the finish option in the part number.

3.3.1.1 Cadmium or Zinc-Nickel Plating

When required by the part number, plating of the outside diameter of CRES sleeve bearings shall be in accordance with AMS-QQ-P-416, Type II, Class 2, or AMS2417, Type 2, Grade A or B, and the applicable AS sheet. The 23-hour bake time of the plating specification shall be waived to prevent over-curing of the liner adhesive. Hydrogen embrittlement testing is not required. Dimensions in AS81934/1 and AS81934/2 apply. For the AMS2417 plating, only Grade B shall be permitted on bearings after June 1, 2016.

3.3.1.2 Passivation

When required by the part number, exposed metallic surfaces of unplated CRES sleeve bearings shall be passivated per AMS2700, Method 1 (Nitric Acid) or Method 2 (Citric Acid), or passivate in accordance with ASTM A967, citric acid passivation, using CITRIC 1, CITRIC 2, or CITRIC 3 solutions after all metal removal operations, regrinding inclusive, have been completed. Passivation may be performed before or after bonding, at the bearing manufacturer's option. A permissible method for determining passivity is the copper sulfate test defined in ASTM A967 or AMS2700.

3.3.2 Aluminum Finishes

Aluminum sleeve bearings shall be anodized per MIL-A-8625, Type I or II, or chemical film treated per MIL-DTL-5541.

3.4 Design

Bearing design shall conform to that shown on AS81934/1 and AS81934/2.

3.5 Construction

The liner shall be so secured that all relative motion will be between the liner and the shaft. Once the liner is installed, a gap of 0.025 inch maximum between the ends of the liner where they meet on the inside diameter of the bearing is allowed. This 0.025 inch maximum gap shall have resin covering the full gap area so no substrate is exposed.

3.5.1 Dimensions and Tolerances

Dimensions and tolerances shall be as specified on the applicable AS sheet. Dimensions not shown shall be at the option of the manufacturer. Lined IDs shall be inspected per ARP5448/9.

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3.5.2 Surface Texture

The surface texture shall be in accordance with the applicable AS sheet and ASME B46.1. Bearings shall be free of any surface defects which may be detrimental to satisfactory installation, performance or bearing life as defined in this standard. Unless otherwise specified, liner surfaces are exempt from surface texture measurements.

3.5.3 Lubrication

Initial grease or oil lubrication by the manufacturer will not be permitted.

3.5.4 PTFE Liner Condition

3.5.4.1 Visual Examination

The visual appearance of the bonded liner shall exhibit a degree of workmanship consistent with proper manufacturing process controls, as checked per 4.6.6.1. The liner shall be uniform in texture and shall contain no embedded contaminants. The liner setback shall meet the applicable drawing requirements. There shall be no separation or lifting of the liner at any of the edges. Unraveling and fraying is not permitted outside the setback area. Molded liners shall not contain embedded contaminants, cracks, or bubbles.

3.5.4.2 Bond Integrity and Peel Strength

When tested in accordance with 4.6.6.2, the liner shall be tightly adherent to the metallic substrate over at least 90% of the contact area and shall exhibit an average peel strength of 2 pounds per inch or greater. The adhesive remaining on the metal substrate shall have no void or unbonded area which cannot be included within a circumscribing circle with a diameter equal to 25% of the lined sleeve length or 0.25 inch, whichever is smaller.

3.5.4.2.1 Processing Controls

All fabrication of the PTFE liner involving application or mixing of adhesive, and all liner bonding procedures involving application of adhesive, shall be conducted in a controlled area.

3.5.4.2.1.1 Controlled Area

The controlled area shall be maintained at a temperature of 75 °F ± 10 °F with a maximum relative humidity of 75%. The enclosed atmosphere of the work area shall be well ventilated and maintained so that the particle count is 2500/ft^3 maximum for particles 5 µm or larger when measured per ASTM F25 or ASTM F50. The particle count measurement shall be performed annually as a minimum. There shall be no eating or smoking in the controlled area and no process which produces uncontrolled spray, dust, fumes, or particulate matter.

3.5.4.3 Liner Peelability

Each manufacturer shall establish during qualification testing whether the liner is peelable or non-peelable. If peelable, the manufacturer shall determine the mean and standard deviation peel strength values for the liner based upon a minimum of six peel strength tests conducted in accordance with 4.6.6.2. Prior to peel testing, three of the six parts must undergo LN_2 exposure per 4.6.7. Liners are to remain bonded after the LN_2 exposure test. This data shall be recorded in the qualification test report. A liner originally qualified as peelable shall remain peelable in production. A liner originally qualified as non-peelable shall remain non-peelable in production.

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3.6 Performance

3.6.1 Radial Static Limit Load

After the radial static limit load listed in Table 1 has been applied as specified in 4.6.1, the permanent set shall not exceed 0.002 inch.

Table 1 - Load values

Part No. 1/	Radial Static Limit Load (pounds)	Oscillation Load (pounds)	Loaded Breakaway Torque 2/ (in-lb max)
M81934/1-08A012	6900	6300	79
M81934/1-08C012	10800	6300	79
M81934/1-16A016	20000	16500	410
M81934/1-16C016	31400	16500	410
M81934/1-24A016	30000	22500	840
M81934/1-24C016	47100	22500	840

1/ Load ratings apply to Type A bearings also.

2/ Loaded breakaway torque is measured after completion of the oscillation under radial load test specified in 4.6.2, using the oscillation load specified in Table 1. The listed loaded breakaway torque values were calculated based on loaded bearings with a coefficient of friction of 0.05.

3.6.2 Oscillation Under Radial Load

The total liner wear of the bearing shall not exceed 0.0035 inch after 1000 cycles, 0.0040 inch after 5000 cycles, and 0.0045 inch after 25000 cycles, when tested at room temperature in accordance with 4.6.2. Additionally, if Type A bearings, the maximum wear limit is 0.0045 inch after 100000 cycles. Visual examination of the liner after test shall indicate no loss of bonding to the metal substrate in the loaded area. Loaded rotational breakaway torque shall not exceed the values in Table 1 after the oscillation under radial load test.

3.6.3 Fluid Compatibility

When tested in accordance with 4.6.3, the bearings shall be compatible with the fluids listed in 4.6.3 and the total liner wear shall not exceed 0.006 inch. Visual examination of the liner after test shall indicate no loss of bonding to the metal substrate in the loaded area.

3.6.4 High Temperature

When tested in accordance with 4.6.4 under the dynamic load specified in Table 1, the total liner wear shall not exceed 0.006 inch. Visual examination of the liner after test shall indicate no loss of bonding to the metal substrate in the loaded area.

3.6.5 Sub-Zero Temperature

When tested in accordance with 4.6.5, the bearings shall be capable of operating at a temperature of -65 °F. The liner wear shall not exceed 0.008 inch. Visual examination of the liner after test shall indicate no loss of bonding to the metal substrate in the loaded area.

3.7 Interchangeability

All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other and with respect to installation and performance. The drawing number requirements of ASME Y14.100 shall govern documentation of and changes in the manufacturer's part numbers. At the discretion of the end user, a Type A bearing listed on the QPL may be substituted for the same non-Type A bearing on the QPL. Non-Type A bearings shall not be substituted for Type A bearings.

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3.8 Identification of Product

For -08 size bore codes and larger (as shown in the applicable AS sheets) – Each bearing shall be permanently and legibly marked with the manufacturer's CAGE code, manufacturer's lot number, and Aerospace Standard part number, as a minimum. Any additional marking is optional. Metal impression stamping is prohibited.

For -07 size bore codes and smaller (as shown in the applicable AS sheets) – Each bearing shall be permanently and legibly marked with the manufacturer's CAGE code and manufacturer's lot number as a minimum. The AS part number is not required to be marked on the part. Manufacturing lot number may be truncated to facilitate marking (i.e., eliminate abbreviation or reference to the words meaning job, lot, lot number, or job number while still including the actual lot number for traceability). A truncated lot number must be readily identified by manufacturer. All -07 size bore codes and smaller manufactured after October 1, 2016, shall have these marking requirements.

3.9 Workmanship

3.9.1 General Requirements

All workmanship shall be such as to result in a first quality bearing which meets the requirements of this standard. The bearing shall be free from defects which may affect its durability and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection

Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. The supplier may use his/her own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the qualifying activity. The qualifying activity reserves the right to perform any of the inspections set forth in this standard where such inspections are deemed necessary to ensure product and/or services conform to prescribed requirements.

4.1.1 Responsibility for Compliance

All items shall meet all requirements of Sections 3, 4, and 5. The inspection set forth in this standard shall become a part of the supplier's overall inspection system or quality program. The absence of any inspection requirements in the standard shall not relieve the supplier of the responsibility of ensuring that all products or services submitted to the qualifying activity for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the customer to accept defective material.

4.1.2 Qualification Test Records

The manufacturer shall maintain a record showing quantitative results for all tests required by this standard for 7 years, or as contractually obligated. This record shall be available to the purchaser and shall be signed by an authorized representative of the manufacturer or the testing laboratory, as applicable.

4.2 Classification of Inspections

The inspection requirements specified herein are classified as follows:

a. Qualification inspection (4.3).

b. Quality conformance inspection (4.4).

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4.3 Qualification Inspection

4.3.1 Sampling Instructions

Qualification test samples shall consist of 37 aluminum and/or CRES -08 size bearings depending upon the substrate(s) to be qualified per Tables 2A or 2B. Additionally, 13 bearings of each of the other sizes and lengths specified in column I of Tables 2A or 2B for which qualification is desired are to be tested (these bearings must be the same material as the qualification bearings, aluminum or CRES). Corresponding bearings in column II of Tables 2A and 2B will be qualified on the basis of similarity. All bearings necessary for tests specified herein shall be furnished by the manufacturer. Samples shall be identified as required and forwarded to the activity designated in the letter of authorization (see 6.3 and 6.3.1).

Table 2A - Order of qualifying bearings

I	II
CRES	
M81934/1-08C012 or M81934/2-08C012	M81934/1 & /2-04C thru -09C
M81934/1-16C016 or M81934/2-16C016	M81934/1 & /2-10C thru -18C
M81934/1-24C016 or M81934/2-24C016	M81934/1 & /2-20C thru -32C
Aluminum	
M81934/1-08A012 or M81934/2-08A012	M81934/1 & /2-04A thru -09A
M81934/1-16A016 or M81934/2-16A016	M81934/1 & /2-10A thru -18A
M81934/1-24A016 or M81934/2-24A016	M81934/1 & /2-20A thru -32A

Table 2B - Order of qualifying bearings for Type A bearings

I	II
CRES	
M81934/1A-08C012 or M81934/2A-08C012	M81934/1A & /2A-04C thru -09C
M81934/1A-16C016 or M81934/2A-16C016	M81934/1A & /2A-10C thru -18C
M81934/1A-24C016 or M81934/2A-24C016	M81934/1A & /2A-20C thru -32C
Aluminum	
M81934/1A-08A012 or M81934/2A-08A012	M81934/1A & /2A-04A thru -09A
M81934/1A-16A016 or M81934/2A-16A016	M81934/1A & /2A-10A thru -18A
M81934/1A-24A016 or M81934/2A-24A016	M81934/1A & /2A-20A thru -32A

4.3.2 Certified Test Report

The manufacturer shall furnish a certified test report showing that the manufacturer's product satisfactorily conforms to this standard (see 6.3.1). The test report shall include, as a minimum, actual results of the tests specified herein. For each dynamic test specified in 3.6.2 through 3.6.5, the reported results shall include loaded breakaway torque values, liner wear curves, and liner bond condition (per 3.6.2). When the report is submitted, it shall be accompanied by a dated drawing which completely describes the manufacturer's product by specifying all dimensions and tolerances, bearing material, coating or plating, and heat treatment. The manufacturer's part number for each size shall be included on the drawing.

4.3.3 Tests

Qualification tests shall include all the examinations and tests of this standard. The minimum number of samples per test shall be in accordance with Table 3.

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There are separate qualification requirements for CRES and aluminum AS81934 bearings; see below:

a. CRES – Qualification to CRES AS89134 bearings can be partially achieved by similarity if the following conditions are all satisfied and certified.

1. If a bearing manufacturer's liner is qualified to AS81820 for sizes MS14101-5A, -8, and -12 or MS14101A-5A, -8, and -12 for Type A.

2. The liner materials and bonding procedures used in these bearings are the same as qualified to AS81820.

If the conditions above are met, qualification for CRES AS81934 bearings shall still require fluid compatibility testing as outlined in 4.6.3 (18 total samples required).

If the manufacturer does not have AS81820 qualification for the liner system, AS89134 qualification requirements are as outlined in Table 3.

b. Aluminum – Qualification for aluminum AS89134 bearings is as outlined in Table 3.

Table 3 - Qualification test samples

Examination and Tests	Paragraph Number	Samples to be Tested Initial Size (-08, preferably)	Samples to be Tested Additional Sizes
Examination of product (non-destructive)	4.5.1	5	5
Preparation for delivery (non-destructive)	4.5.2	5	5
Radial static limit load	4.6.1	3	3
Oscillation under radial load	4.6.2	3	3
Fluid compatibility 1/	4.6.3	18	-
High temperature 2/	4.6.4	3	-
Sub-zero temperature 2/	4.6.5	3	-
Bond integrity	4.6.6.2	1	1
Peel strength 3/	4.6.6.2	6	6

1/ Three for each fluid (fluid compatibility testing is required for one size range only (preferably -08).
2/ Three each for one size range only (preferably -08).
3/ Peel strength applies to peelable liners only.

4.3.4 Retention of Qualification

The continued listing of a product on the Qualified Products List is dependent upon a periodic verification of the manufacturer's continued compliance with the requirements of this document and with standardization regulations. As part of that verification process, each manufacturer must complete DD Form 1718 every 2 years. This form, supplied by the NAWCAD Joint Base MDL, NJ, is to be signed by a responsible official of management and sent to Commander, Naval Air Warfare Center Aircraft Division, Code 412000B120-3: Joint Base MDL, NJ 08733-5100.

4.3.5 Periodic Retest for Retention of Qualification

In addition to the certification requirement of 4.3.4, each manufacturer shall retest a sample of bearings from recent production at 4 year intervals beginning in 2016. The tests to be performed shall be as follows:

a. Oscillation under radial load (4.6.2).

b. Fluid compatibility (4.6.3), using only one contamination fluid from 4.6.3 (to be specified by the qualifying activity).

c. High temperature (4.6.4).

d. PTFE liner condition (4.6.6).

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The test bearings shall be a single size from normal production and shall be the currently qualified bearing material (aluminum and/or CRES). Three samples each of test (a), (b), and (c), and seven samples of test (d) are required. The sample bearings shall be of the -08 size specified in column I of Tables 2A or 2B.The test data shall be assembled in report form, certified by the Chief Engineer or other responsible official of management, and forwarded to the qualifying activity in October of the appropriate year. For tests (a), (b), and (c), the reported results shall include, as a minimum, the loaded breakaway torque values, liner wear curves, and liner bond condition (per 3.6.2). If the bearing manufacturer is qualified to AS81820 for sizes MS14101-5A, -8, and -12, or MS14101A-5A, -8, or -12 for Type A, and the liner material and bonding procedures used in these sleeve bearings are the same as qualified to AS81820, the periodic retest for retention of qualification requirements of AS81820 will satisfy the requirements of this paragraph for CRES substrates. If qualified for an aluminum substrate, it is required to periodically retest M81934/1(A)-08A012 bearings during the normal AS81934 retention of qualification period. The tests described above in (a) through (d) are required for the aluminum substrate.

4.4 Quality Conformance

The quality conformance inspections of the bearings shall consist of the examinations and tests of Table 4 to determine conformance of the bearings to the requirements of this standard.

Table 4 - Quality conformance inspections

Examination or Test	Major Characteristics	Minor Characteristics	Special Inspection Plan	Requirement Paragraph	Test Paragraph
(a) Dimensions/AS Ref					
Bore "B"	X			3.5.1	4.5.1
Sleeve Diameter "D"	X			3.5.1	4.5.1
Sleeve Diameter "H" 1/		X		3.5.1	4.5.1
Flange Thickness "F" 1/	X			3.5.1	4.5.1
Length "L"		X		3.5.1	4.5.1
Concentricity	X			3.5.1	4.5.1
Chamfers		X		3.5.1	4.5.1
Liner Setbacks		X		3.5.1	4.5.1
Undercut Features 1/		X		3.5.1	4.5.1
(b) Identification of Product		X		3.8	4.5.1
(c) Workmanship		X		3.9	4.5.1
(d) Preparation for Delivery		X			4.5.2
(e) PTFE Liner Condition 2/			Destructive Inspection	3.5.4	4.6.6

1/ Inspection applies only to M81934/2 or M81934/2A for Type A parts.
2/ The LN_2 exposure requirement does not apply to quality conformance inspections of production lots.

4.4.1 Inspection Lot

The inspection lot shall consist of finished bearings, having a single part number, manufactured according to same procedures as the parts originally qualified and produced as one continuous run or order or portion thereof.

4.4.2 Sampling

4.4.2.1 Sample for Quality Conformance Tests in Table 4 (a) through (d)

The sample bearings shall be selected at random from each inspection lot in accordance with Table 5. If no defect is found in the sample lot, the lot shall be accepted for these tests. If any defects are found in the sample, the entire lot shall be 100% inspected for each defective characteristic found, and all defective parts shall be removed from the lot.

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Table 5 - Sampling plan for quality conformance inspections except destructive inspections zero-based acceptance plan (C = 0; i.e., no rejects)

Lot Size	Major Characteristics	Minor Characteristics
1-2	A	A
3-8	A	3
9-12	A	3
13-15	13	3
16-25	13	3
26-50	13	5
51-90	13	6
91-150	13	7
151-280	20	10
281-500	29	11
501-1200	34	15
1201-3200	42	18
3201-10000	50	22

4.4.2.2 Sample for Quality Conformance Test in Table 4 (e)

The sample bearings shall be selected at random from each inspection lot in accordance with Table 6. At the bearing manufacturer's option, the sample bearings for destructive inspections may be selected at random from the non-repairable defective parts rejected from the inspection lot during quality conformance tests (a) through (d). If no defect is found in the sample, the lot shall be accepted for these tests. A lot which was rejected under normal inspection of Table 6 shall be reinspected under tightened inspection of Table 6. If a defect is found in the tightened inspection sample, the entire lot shall be rejected and shall not be offered for sale or acceptance under this standard.

Table 6 - Destructive inspection sampling plan zero-based acceptance plan (C = 0; i.e., no rejects)

	Normal Inspection			Tightened Inspection		
Lot Size	Sample Size	Accept	Reject	Sample Size	Accept	Reject
2-50	2	0	1	4	0	1
51-500	3	0	1	6	0	1
501-5000	5	0	1	10	0	1
5001-50000	8	0	1	16	0	1

4.4.3 Quality Assurance Certification

For each inspection lot the manufacturer shall maintain and supply to the customer upon demand:

a. Certified copies of all records of quality conformance tests specified in 4.4 and the purchase order.

b. Certification that the materials, manufacturing procedures, and processes used in producing the bearings are the same as those of the bearings originally qualified.

These records and certifications shall be kept for 7 years or as contractually obligated.

4.4.4 Resubmitted Inspection Lots

Where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be selected. A resubmitted inspection lot shall be inspected using tightened inspection. When an inspection lot is resubmitted, full particulars concerning the cause of previous rejection and the action taken to correct the defects found in the inspection lot, shall be furnished by the contractor to the procuring activity.

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4.5 Examinations

4.5.1 Examination of Product

The bearings shall be examined to determine conformance to the requirements of this standard and the applicable AS sheet for material, dimensions, finish, identification of product, workmanship, and requirements not covered by tests.

4.5.2 Preparation for Delivery

Preservation, packaging, packing, and marking shall be inspected to determine conformance to Section 5.

4.6 Test Methods

Unless otherwise specified, all tests shall be conducted at room temperature.

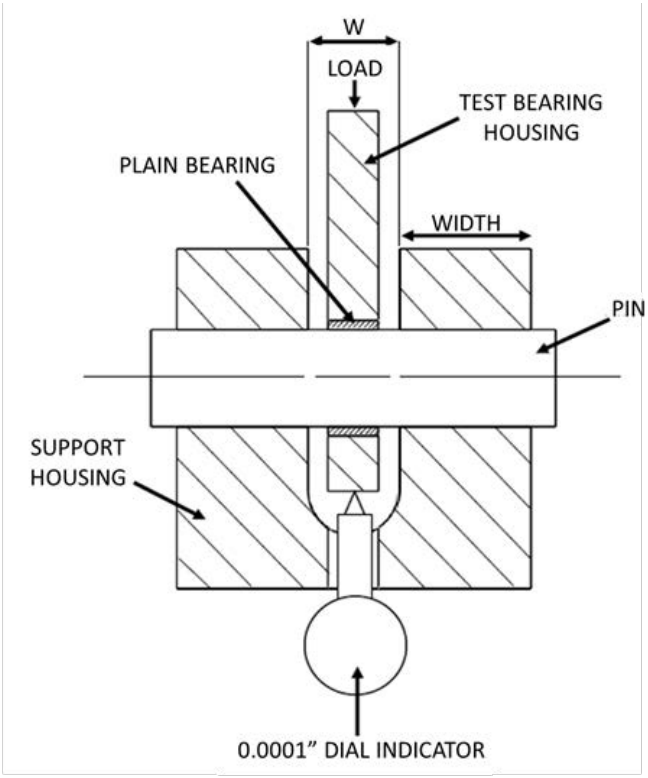
4.6.1 Radial Static Limit Load

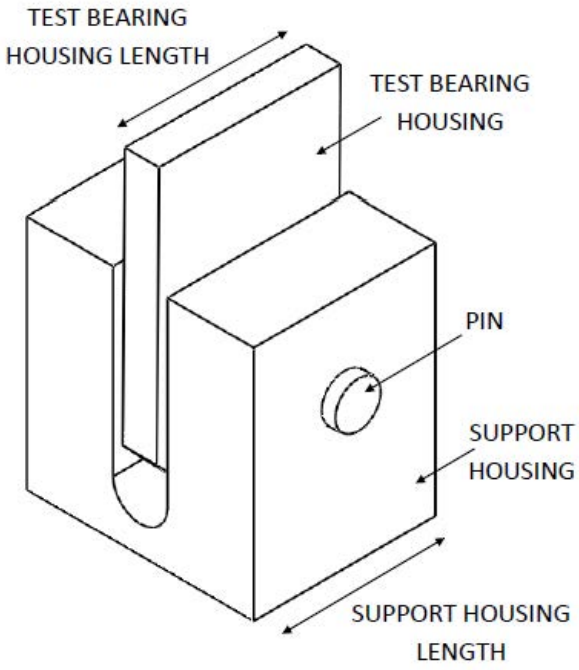
The bearings shall be installed in a test fixture as shown on Figure 1. A preload of 4 to 6% of the radial static limit load shall be applied to the bearing for 3 minutes, and then the displacement measuring device should be set at zero. The load shall then be increased at the rate of 1% of the specified load per second until it equals the radial static limit load listed in Table 1. The load shall then be reduced at the same rate to the preload value. The permanent set shall be the reading at preload (see 3.6.1).

4.6.2 Oscillation Under Radial Load

At any time prior to installation in the test housing, each sleeve bearing is to be exposed to LN_2 per 4.6.7 (only one LN_2 exposure is required for any qualification test). The bearing shall be installed in a steel housing using an interference fit with the housing and a loose fit with the pin, with housing and pin tolerances per Figure 1. The bearing shall be so installed as to place the pin in double shear. A dial indicator or electronic pickup shall be so mounted that any radial movement of the pin or the bore of the bearing with respect to the bearing outside diameter can be measured. The oscillation load specified in Table 1 shall be applied and held statically for 15 minutes. At the end of this time, the indicating device shall be set at zero and the oscillating test shall be started. Wear readings shall include the wear from the first cycle on. The test shall be run in such a manner that the pin is oscillated at ±25 degrees from the zero position (50 degree included angle, or 100 degree total travel per cycle) a minimum of 10 cpm for 25000 cycles (or 100000 cycles for Type A liner system). One cycle shall consist of rotation from 0 to +25 degrees, return through 0 to -25 degrees, and return to 0 degrees. Sufficient readings during the test shall be recorded to plot wear in thousandths of an inch versus life in cycles. Upon completion of the test, the loaded rotational breakaway torque shall be as specified in Table 1 and liner wear and liner bond shall be as specified in 3.6.2.

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Pin:

Diameter = B_{min} +0.000/-0.002 inch
Material = Steel 50 Rc min
Finish = 8 Ra max
Pin Taper: Taper at least one pin end for installation; pin tapers are not allowed within the specified support housing widths when assembled.

Test Bearing Housing:

Bore = D – (0.0006 to 0.0011 inch)
Width = L_{max} +0.002/-0.000 inch
Length (min.) = 2 X D
Material = Steel

Support Housing:

Bore = (B_{min} + 0.001) + (0.0002 to 0.0027 inch)
Width (min.) = L
Length (min.) = 2 X B
For unflanged test bushings: W = L_{max} + (0.025 to 0.030 inch)
For flanged test bushings: W1 = F + L_{max} + (0.025 to 0.030 inch)
Material = Steel

NOTE: L, B, F, and D indicate nominal dimensions from the applicable AS sheet. Max and min dimension modifiers indicate the extremes of the allowable tolerance ranges listed in the applicable AS sheet.

Figure 1 - Radial test fixture

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4.6.3 Fluid Compatibility

Prior to being immersed in any contamination fluid, each bearing is to be exposed to LN_2 per 4.6.7.

Eighteen bearings (three selected for each fluid per 4.3.3) shall be immersed for 24 hours in each of the following fluids at 160 °F ± 5 °F, except for (b), which shall be at 110 °F ± 5 °F:

a. Phosphate Ester hydraulic oil per AS1241, Type IV or Type V.

b. Turbine fuel grades JP-5 per MlL-DTL-5624, JP-8 (NATO F-34) per MIL-DTL-83133, or NATO F-24 (Jet A per ASTM D1655 with the military additives SDA, S-1745 (FSII), and S-1747 (CI/LI)).

c. MIL-PRF-7808 lubricating oil.

d. MIL- PRF-5606 hydraulic oils.

e. Propylene-glycol based AMS1424 anti-icing fluid, diluted with water to achieve a freezing point of -30 °F ± 10 °F (approximately 50:50 by volume for most AMS1424 concentrates) as verified by glycol refractometer.

f. MIL-PRF-83282 hydraulic fluid.

NOTE: (e) should be heated in a covered container to reduce the increase in concentration due to evaporation.

Within 1/2 hour after removal from the test fluid and without removing fluid from the bore surface or further immersing the bearing in LN_2 or any other liquid, the bearing shall be tested in accordance with 3.6.3 and 4.6.2. The load for (e) shall be 75% of the oscillation load specified in Table 1.

4.6.4 High Temperature

Prior to installation in the test housing, each bearing is to be exposed to LN_2 per 4.6.7.

Three bearings selected per 4.3.3 shall be subjected to the test of 4.6.2, with the bearings heated in such a way that it has been demonstrated that all locations on the bearing and on the length of the test pin contained by the bearing are continuously maintained at a minimum temperature of 325 °F during the test. At least one primary temperature monitoring location shall be located on or within the bearing test fixture housing, the test bearing, or the test pin; and be located within 0.125 inch distance from any point on pin/liner interface, and within ±45 circumferential degrees from the center of the test bearing's load zone. The temperature sensor shall be shielded or located such that direct exposure to the heating medium does not bias the temperature measurement. The primary temperature monitoring location shall be verified to be 325 °F minimum for the duration of the test. Before applying the oscillation load, the primary temperature monitoring location shall be held at its setpoint temperature long enough achieve thermal equilibrium (stabilized expansion of test fixtures). The test load shall be 100% of the oscillation load specified in Table 1 (see 3.6.4).

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4.6.5 Sub-Zero Temperature

Prior to installation in the test housing, each sleeve bearing is to be exposed to LN_2 per 4.6.7.

Three bearings selected per 4.3.3 shall be subjected to the test of 4.6.2, with the bearings cooled in such a way that it has been demonstrated that all locations on the bearing and on the length of the test pin contained by the bearing are continuously maintained at a maximum temperature of -65 °F during the low temperature portion of this test (see 3.6.5). At least one primary temperature monitoring location shall be located on or within the bearing test fixture housing, the test bearing, or the test pin; and be located within 0.125 inch distance from any point on pin/liner interface, and within ±45 circumferential degrees from the center of the test bearing's load zone. The temperature sensor shall be shielded or located such that direct exposure to the cooling medium does not bias the temperature measurement. The test load shall be 75% of the load specified in Table 1. The oscillation rate is optional except that the minimum shall be 5 cpm. Intermittent operation of the test machine is allowable if necessary to counteract the effects of frictional heating. The first 1000 test cycles shall be conducted at room temperature followed by 24000 cycles (or 99000 cycles for Type A) at or below -65 °F. Upon completion of the first 1000 cycles, oscillation shall be stopped and the test bearing cooled to -65 °F. The primary temperature monitoring location shall be held at its setpoint temperature long enough achieve thermal equilibrium (for a minimum of one-half hour) after which time oscillation shall be continued until the remaining 24000 cycles (or 99000 cycles for Type A) have been completed. The primary temperature monitoring location shall be verified to be -65 °F maximum during the low temperature portion of the test. In the event that the normal wear measurement techniques are inaccurate at low temperature, then alternative wear methods may be used provided the method is fully described in the qualification test report. After testing, one bearing shall be subjected to the bond integrity test of 4.6.6. Successful testing at temperatures lower than -65 °F, or with the first 1000 cycles at low temperature, or at loads higher than 75% are not causes for rejection.

4.6.6 PTFE Liner Condition

4.6.6.1 Visual Examination

Visually examine the exposed liner surface for conformance to the applicable requirements of 3.5.4.1, 3.9, and the AS sheet.

4.6.6.2 Bond Integrity and Peel Strength

For qualification testing only: Prior to peel testing, three of the six parts must undergo LN_2 exposure per 4.6.7. Non-peelable liners to remain bonded after LN_2 test. Peel the liner away from the metal substrate and evaluate the peel strength and adhesive bond appearance per 3.5.4.2. A blade or scribe may be used to initiate the peel. To determine peel strength, the liner shall be attached to a calibrated spring scale or tension testing machine of the type described in ASTM C794 and the liner peeled back from the substrate at 140 degree ± 40 degree angle to the bond surface at a speed of 0.5 to 1 in/min (see Figure 2). The average peel strength value shall be recorded and included in the test report. Where possible, the peel shall be conducted on the entire width of the bonded liner. When the peel is conducted on the entire width of the sleeve, the width for calculating peel strength in pounds per inch shall be the machined width minus 0.050 inch to account for chamfers and edge effects. At the manufacturer’s option, or when peeling of the full sleeve width is impractical, the liner may be cut through to the metal substrate to form a peel sample with a minimum width of 0.500 inch. This peel sample may be cut either parallel to or perpendicular to the sleeve side face. This sample shall be peeled as previously described. The peel strength test shall be waived for the -04 through -09 size bearings, but the liner shall be peeled by hand to permit visual examination of the bond line for voids.

Following liner peel, the adhesive bond appearance shall be evaluated to determine the location and size of any voids or unbonded areas as specified in 3.5.4.2. In the event the liner cannot be removed without employing chipping, scraping, or abrasive techniques, the liner shall be considered to be properly bonded and free of voids and shall be classed as non-peelable. Non-peelable liners are not required to meet the peel strength test.

4.6.7 Liquid Nitrogen (LN_2) exposure

When LN_2 exposure is specified, the bearing shall be submerged in LN_2 for a minimum of 5 minutes. Ensure that the bearing is completely dry before submerging. There must be sufficient LN_2 in the container to keep the bearing completely submerged for the full 5 minutes. Using tongs or similar suitable device, remove the bearing from the LN_2 and allow the bearing to come to 65 to 85 °F prior to beginning the applicable test.

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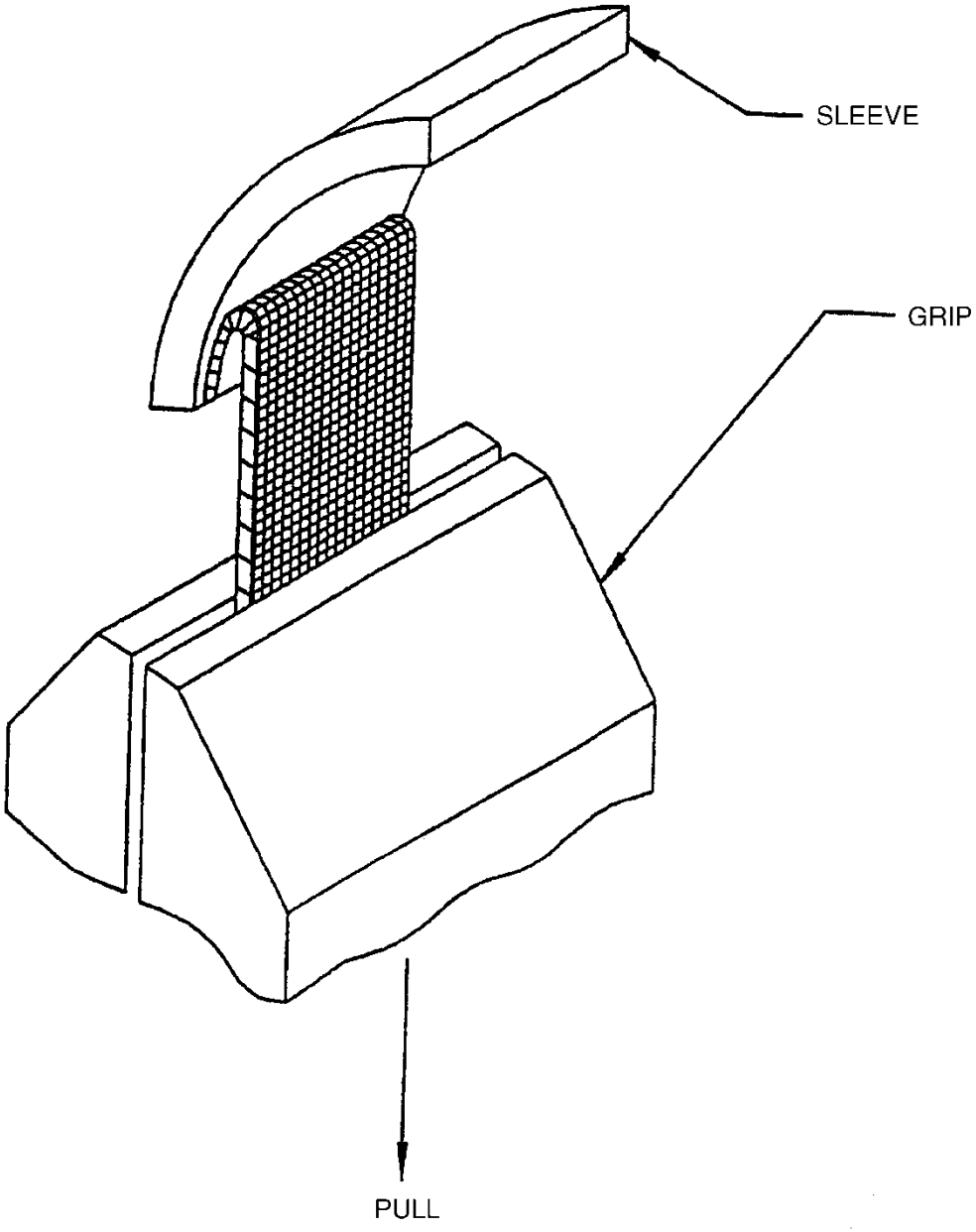


Figure 2 - Example of peel test fixture

5. PACKAGING

5.1 Preservation

Preservation shall be in accordance with the requirements of MIL-DTL-197.

5.1.1 Level of Preservation

The military level of preservation shall be that determined by the purchasing document or contract and as described in Section 3 of MIL-DTL-197.

5.1.2 Method of Preservation

The method for preservation shall be as described in Section 3 of MIL-DTL-197 or as modified in the purchasing document or contract.

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5.1.3 Cleaning

The method of cleaning shall be the manufacturer's established cleaning procedures for this bearing. MIL-DTL-197 describes several cleaning processes and limitations for use of solvents, greases, and oils on self-lubricating bearings.

5.2 Packing

Packing shall be in accordance with MIL-DTL-197 and MIL-STD-2073-1. Commercial shall be the normal level of packing. The acquisition document or contract may be modified by the acquisition officer to specify Level A or Level B packing as required. The detailed requirements for the packing, packaging, containers, cushioning, etc., are as described in Sections 3 and 4 of MIL-DTL-197 and in MIL-STD-2073-1.

5.3 Marking

Marking of unit, intermediate, and exterior containers shall be in accordance with MIL-STD-129. The use of additional markings required by specifications, drawings, and customer requirements are authorized.

5.4 Commercial Packaging

Commercial preservation and packaging requirements have been deleted from MIL-DTL-197. ASTM D3951 is a guide for commercial packaging. Note that clean room requirements and other requirements found in MIL-DTL-197 are not required when using commercial processing.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

NOTICE

This document references a part which contains cadmium as a plating material. Consult local officials if you have questions concerning cadmium's use.

6.1 Intended Use

These bearings are intended primarily for use in airframe applications of high loads at low rotational oscillatory speeds. For specific design information on the capability of these bearings under particular load speed and wear/life conditions, the user is referred to MIL-HDBK-1599.

6.2 Acquisition Requirements

Acquisition documents must specify the following:

a. Title, number, and date of this standard.

b. Type and complete AS number.

c. Quantity.

d. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

e. Quality assurance certification (see 4.4.2).

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6.3 Qualification

With respect to products requiring qualification, contract awards will be made only for such products having been approved for inclusion in Qualified Products List (QPL-AS81934) and Qualified Products Database, whether or not such products have actually been so listed by that date. An online listing of products may be found in the Qualified Products Database (QPD) at https://assist.dla.mil/, https://qpldocs.dla.mil/, or https://quicksearch.dla.mil/. The qualifying activity is the Naval Air Systems Command and information pertaining to qualification of products may be obtained from that command at AIR VEHICLE ENGINEERING DEPARTMENT, ATTN: CODE 4.3.5.4 BLDG 2187, NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION, 48110 SHAW ROAD STE 3320, PATUXENT RIVER MD 20670-1907. Manufacturers should make a request to the qualification activity for authorization to begin their qualification test program.

6.3.1 Authorization for Submittal of Samples

A manufacturer seeking qualification approval of his/her product will be authorized to submit samples for such approval only upon presentation of certified test reports and drawings indicating that his/her product conforms to this standard.

6.4 Plating Process Caution

Caution should be taken during plating process. The supplier should be responsible for the safe reutilization and disposal of all material generated by this process in accordance with ASTM A380, Sections 8.2 and 8.7.

6.5 Revision Indicator

A change bar (l) located in the left margin is for the convenience of the user in locating areas where technical revisions, not editorial changes, have been made to the previous issue of this document. An (R) symbol to the left of the document title indicates a complete revision of the document, including technical revisions. Change bars and (R) are not used in original publications, nor in documents that contain editorial changes only.

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